Spectrum
SPC

From Binary to Beyond

بنك أسئلة C++

66 سؤال اختيار من متعدد · C++ Midterm Practice

@spc.bau

كل البرامج تبدأ بالترويسات التالية، واختر ناتج كل برنامج:

```
#include <iostream>
#include <cmath>
#include <string>
#include <cstring>
#include <iomanip>
using namespace std;
```

المستوى الأول

1 What is the output of the following code?

```
int main() {
    bool b = true;
    cout << b;
}
```

A 1
B b
C true
D 0

2 What is the output of the following code?

```
int main() {
    int a = 10, b = 20;
    int t = a;
    a = b;
    b = t;
    cout << a << " " << b;
}
```

A 20 20
B 10 10
C 10 20
D 20 10

3 What is the output of the following code?

```
int main() {
    int x = 7;
    cout << (x % 2 == 0 ? "Even" : "Odd");
}
```

A Odd
B Even
C 1
D 0

4 What is the output of the following code?

```
int main() {
    int x = 0;
    if (x)
        cout << "T";
    else
        cout << "F";
}
```

A Compilation error
B F
C 0
D T

5 What is the output of the following code?

```
int main() {
    int a = 5;
    cout << a++;
}
```

A 5
B 7
C 4
D 6

6 What is the output of the following code?

```
int main() {
    int x = 17;
    x %= 5;
    cout << x;
}
```

A 3
B 2
C 0
D 12

7 What is the output of the following code?

```
int main() {
    cout << (5 > 2 && 3 > 4);
}
```

A 0
B true
C false
D 1

8 What is the output of the following code?

```
int main() {
    int x = 10;
    cout << x / 3;
}
```

A 4
B 3
C 1
D 3.33333

9 What is the output of the following code?

```
int main() {
    cout << sizeof(char);
}
```

A 8
B 4
C 1
D 2

10 What is the output of the following code?

```
int main() {
    cout << 10 % 3;
}
```

A 1
B 0
C 3
D 3.33

11 What is the output of the following code?

```
int main() {
    int a = 3, b = 4;
    cout << (a < b);
}
```

A 1
B 0
C true
D 3

12 What is the output of the following code?

```
int main() {
    int x = 5;
    if (x > 3)
         cout << "Yes";
    else
         cout << "No";
}
```

A No
B Compilation error
C YesNo
D Yes

13 What is the output of the following code?

```
int main() {
    double d = 7 / 2.0;
    cout << d;
}
```

A 4
B 3
C 3.5
D 3.50

14 What is the output of the following code?

```
int main() {
    int x = 10;
    if (x == 10)
        cout << "A";
    cout << "B";
}
```

A A
B B
C AB
D Compilation error

15 What is the output of the following code?

```
int main() {
    cout << "Sum=" << 2 + 3;
}
```

A Compilation error
B Sum=5
C Sum=23
D Sum=2+3

16 What is the output of the following code?

```
int main() {
    char c = 'a';
    cout << (char)(c + 1);
}
```

A 98
B a
C c
D b

17 What is the output of the following code?

```
int main() {
    int x = 5;
    x += 3;
    cout << x;
}
```

A 5
B 15
C 53
D 8

18 What is the output of the following code?

```
int main() {
    cout << (10 - 2 - 3);
}
```

A 5
B 11
C 15
D 9

19 What is the output of the following code?

```
int main() {
    cout << !(5 > 2);
}
```

A 5
B 1
C true
D 0

20 What is the output of the following code?

```
int main() {
    int x = 3 + 4 * 2 - 1;
    cout << x;
}
```

A 12
B 13
C 10
D 8

21 What is the output of the following code?

```
int main() {
    int x = 5;
    if (x > 10)
         cout << "A";
    else if (x > 3)
         cout << "B";
    else
         cout << "C";
}
```

A AB
B B
C C
D A

22 What is the output of the following code?

```
int main() {
    int a = 5;
    a++;
    cout << a;
}
```

A 5
B 7
C Compilation error
D 6

23 What is the output of the following code?

```
int main() {
    int a = 5;
    cout << "a";
}
```

A Compilation error
B 5
C a
D a=5

24 What is the output of the following code?

```
int main() {
    char c = 'A';
    cout << (int)c;
}
```

A A
B 65
C 64
D 97

25 What is the output of the following code?

```
int main() {
    cout << 10 / 4.0;
}
```

A 2.50
B 2
C 2.5
D 3

26 What is the output of the following code?

```
int main() {
    int a = 5;
    cout << ++a;
}
```

A 6
B 5
C 7
D 4

27 What is the output of the following code?

```
int main() {
    cout << 5 + 3 * 2;
}
```

A 11
B 10
C 13
D 16

28 What is the output of the following code?

```
int main() {
    cout << 2 * 3 + 4 * 5;
}
```

A 26
B 50
C 46
D 30

29 What is the output of the following code?

```
int main() {
    int x = 10;
    x -= 4;
    x *= 2;
    cout << x;
}
```

A 12
B 8
C 20
D 6

30 What is the output of the following code?

```
int main() {
    float f = 3.7;
    int n = f;
    cout << n;
}
```

A 3.7
B 4
C 3.0
D 3

31 What is the output of the following code?

```
int main() {
    cout << 7 / 2 * 2.0;
}
```

A 6
B 7
C 6.5
D 3

32 What is the output of the following code?

```
int main() {
    int a = 8;
    cout << a / 3 * 3;
}
```

A 8
B 6
C 2
D 9

33 What is the output of the following code?

```
int main() {
    int a = 7, b = 2;
    cout << a / b << " " << a % b;
}
```

A 3 0
B 3.5 1
C 3 1
D 4 1

34 What is the output of the following code?

```
int main() {
    int x = 2;
    x = x * x + 1;
    cout << x;
}
```

A 5
B 6
C 4
D 3

35 What is the output of the following code?

```
int main() {
    cout << "A" << "B" << endl << "C";
}
```

A A
B
C
B ABC
C AB
C
D AB C

36 What is the output of the following code?

```
int main() {
    cout << (5 > 2 || 3 > 4);
}
```

A false
B 0
C 1
D true

37 What is the output of the following code?

```
int main() {
    double d = 7 / 2;
    cout << d;
}
```

A 3
B 4
C 3.0
D 3.5

المستوى الثاني

38 What is the output of the following code?

```
int main() {
    int a = -7, b = 3;
    cout << a % b << " " << a / b << " ";
    cout << -a % b << " " << -a / b;
}
```

A -1 -3 1 3
B -1 -2 1 2
C -1 -2 -1 -2
D 2 -3 1 2

39 What is the output of the following code?

```
int main() {
    int a = 4;
    int b = a-- * 2;
    int c = --a + b;
    cout << a << " " << b << " " << c;
}
```

A 2 8 10
B 3 8 11
C 2 8 12
D 3 8 10

40 What is the output of the following code?

```
int main() {
    double r = pow(2, 3);
    double s = sqrt(16);
    cout << r + s << " " << r * s << " ";
    cout << abs(-3) + pow(3, 2);
}
```

A 12 16 12
B 10 32 12
C 12 32 12
D 12 32 9

41 What is the output of the following code?

```
int main() {
    double pi = 3.14159;
    cout << fixed << setprecision(2) << pi << " " << pi * 2;
    cout << " " << setprecision(1) << pi / 2;
}
```

A 3.14 6.28 1.57
B 3.14 6.28 1.6
C 3.14 6.28 1.5
D 3.1 6.3 1.6

42 What is the output of the following code?

```
int main() {
    cout << 'z' - 'a' << " ";
    cout << (char)('a' + 3) << " ";
    cout << 'd' - 'a' + 1;
}
```

A 25 d 3
B 25 d 4
C 26 d 4
D 122 d 4

43 What is the output of the following code?

```
int main() {
    cout << '5' + 1 << " ";
    cout << (char)('5' + 1) << " ";
    cout << '5' - '0' + 1;
}
```

A 54 6 6
B 6 6 6
C 54 54 6
D 54 6 5

44 What is the output of the following code?

```
int main() {
    int x = 0;
    if (x != 0 && 10 / x > 1)
         cout << "A";
    else
         cout << "B";
}
```

A A
B AB
C Runtime error
D B

45 What is the output of the following code?

```
int main() {
    int x = 5, y = 2;
    int z = x % y * 3 + x / y * 2;
    cout << x / y * y + x % y << " " << z;
}
```

A 4 7
B 5 8
C 5 7
D 6 7

46 What is the output of the following code?

```
int main() {
    double a = 5 / 2.0 * 2;
    int b = 5 / 2 * 2;
    double c = 5 / 2 * 2.0;
    cout << a << " " << b << " " << c;
}
```

A 5 5 4
B 5 4 4
C 5 4 5
D 2.5 4 4

47 What is the output of the following code?

```
int main() {
    int a = 5, b = 0, c = 0;
    if (a > 3 || ++b)
        cout << "T";
    if (a < 3 && ++c)
        cout << "F";
    if (a > 3 && ++c)
        cout << "U";
    cout << b << c;
}
```

A TU01
B T01
C TU00
D TU11

48 What is the output of the following code?

```
int main() {
    cout << fixed << setprecision(2) << 7.0 / 2 << " ";
    cout << 7 / 2 << " " << 10.0 / 3;
}
```

A 3.50 3.50 3.33
B 3.50 3 3.00
C 3.50 4 3.33
D 3.50 3 3.33

49 What is the output of the following code?

```
int main() {
    int a = -5, b = 3;
    cout << abs(a) + abs(b) << " ";
    cout << abs(a + b) << " " << abs(a) * abs(b);
}
```

A 8 8 15
B 8 2 15
C 2 2 15
D 8 2 -15

50 What is the output of the following code?

```
int main() {
    cout << 2 + 3 % 2 * 4 << " ";
    cout << (2 + 3) % 2 * 4 << " ";
    cout << 2 + 3 % (2 * 4);
}
```

A 6 4 6
B 6 4 5
C 10 4 5
D 6 6 5

51 What is the output of the following code?

```
int main() {
    int a = 3;
    int b = a++ + 2;
    int c = ++a * 2;
    cout << a << " " << b << " " << c;
}
```

A 4 5 10
B 5 5 10
C 5 5 8
D 5 5 12

52 What is the output of the following code?

```
int main() {
    int a = 2, b = 3;
    a *= a + 1;
    b += a * 2;
    b %= 5;
    cout << a << " " << b;
}
```

A 6 15
B 3 0
C 6 5
D 6 0

53 What is the output of the following code?

```
int main() {
    int x = 5, y = 3;
    if (x = 0)
         cout << "A";
    else
         cout << "B";
    if (y = x)
         cout << "C";
    else
         cout << "D";
    cout << x << y;
}
```

A AC05
B BD53
C BC03
D BD00

54 What is the output of the following code?

```
int main() {
    cout << (7 > 3 > 1) << (2 < 5 < 3);
    cout << (1 < 2 < 3) << (3 > 2 > 1);
}
```

A 0110
B 0101
C 1010
D 0111

55 What is the output of the following code?

```
int main() {
    int a = 5, b = 10;
    a = a + b;
    b = a - b;
    a = a - b;
    int c = a * b;
    cout << a << " " << b << " " << c;
}
```

A 5 10 50
B 10 5 15
C 10 5 50
D 15 5 75

56 What is the output of the following code?

```
int main() {
    cout << 10 / 3 * 3.0 << " ";
    cout << 10 / 3.0 * 3 << " ";
    cout << 10 % 3 * 3.0;
}
```

A 9 10 3
B 9 9 3
C 9 10 1
D 10 10 3

57 What is the output of the following code?

```
int main() {
    int a = 1, b = 0, c = 1;
    if (a)
         if (b)
              cout << "X";
         else
              cout << "Y";
    if (c)
         cout << "Z";
    else
         cout << "W";
}
```

A Y
B YZ
C XZ
D YW

58 What is the output of the following code?

```
int main() {
    cout << (5 / 2) * (2.0 / 4) << " ";
    cout << 5 / 2.0 * (2 / 4) << " ";
    cout << (5 / 2.0) * (2.0 / 4);
}
```

A 1 0.5 1.25
B 0 0 1.25
C 1 0 1.25
D 1 1 1.25

59 What is the output of the following code?

```
int main() {
    bool a = true, b = false;
    cout << (a || b && b) << ((a || b) && b);
    cout << (!a || b) << (!(a && b));
}
```

A 1011
B 1001
C 1101
D 0001

60 What is the output of the following code?

```
int main() {
    cout << (char)('A' + ('z' - 'a') % 26);
    cout << (char)('a' + ('z' - 'a' + 3) % 26);
}
```

A Zb
B Yc
C Zd
D Zc

61 What is the output of the following code?

```
int main() {
    int a = 2, b = 3, c = 4;
    int d = (a + b) * c - a / b + c % b;
    cout << a + b * c - a / b + c % b << " " << d;
}
```

A 15 20
B 15 21
C 14 21
D 21 15

62 What is the output of the following code?

```
int main() {
    int a = 17, b = 5;
    double c = a / b;
    double d = (double)a / b;
    int e = a % b;
    char f = 'A' + e;
    a += e;
    b *= 2;
    cout << c << " " << d << " " << e << " ";
    cout << f << " " << a << " " << b << endl;
    return 0;
}
```

A 3 3.4 2 B 19 10

B 3 3.4 2 C 19 10

C 3.4 3.4 2 C 19 10

D 3 3.4 3 C 17 10

63 What is the output of the following code?

```
int main() {
    int x = 5;
    int y = x++ * 2;
    int z = ++x * 2;
    x -= y;
    y = y % 4 + z / 5;
    bool flag = (x < 0);
    cout << x << " " << y << " " << z << " " << flag << endl;
    return 0;
}
```

A -3 6 14 1

B -1 4 12 1

C -3 4 14 0

D -3 4 14 1

64 What is the output of the following code?

```
int main() {
    int a = 12, b = 7, c = 20;
    int score = 0;
    if (a > b && b > c) {
        score += 1;
        cout << "A ";
    } else if (a > b || b > c) {
        score += 2;
        cout << "B ";
        if (c % a == 8)
              cout << "C ";
        else
              cout << "D ";
    } else {
        cout << "E ";
    }
    if (!(a < b) && c > a)
        score += 5;
    cout << "F " << score << endl;
    return 0;
}
```

A BCF7
B BCF8
C BDF7
D BCF2

65 What is the output of the following code?

```
int main() {
    int a = 0, b = 0, c = 0;
    if (a++ > 0 && b++ > 0)
        c = 1;
    cout << a << b << c << " ";
    if (a++ > 0 || b++ > 0)
        c += 2;
    cout << a << b << c << " ";
    if (++a > 5 || ++b > 0)
        c += 4;
    cout << a << b << c << endl;
    return 0;
}
```

A 100 202 306
B 100 202 316
C 110 202 417
D 100 201 316

66 What is the output of the following code?

```
int main() {
    int a = 6, b = 4, c = 2;
    a += b * c;
    cout << a << " ";
    b -= a % 5;
    cout << b << " ";
    c *= a - b;
    cout << c << " ";
    a /= c - 25;
    cout << a << endl;
    return 0;
}
```

A 14 0 28 4

B 14 0 28 5

C 14 4 20 4

D 14 0 14 14

مفتاح الإجابات

1	A	2	D	3	A	4	B	5	A	6	B
7	A	8	B	9	C	10	A	11	A	12	D
13	C	14	C	15	B	16	D	17	D	18	A
19	D	20	C	21	B	22	D	23	C	24	B
25	C	26	A	27	A	28	A	29	A	30	D
31	A	32	B	33	C	34	A	35	C	36	C
37	A	38	B	39	A	40	C	41	B	42	B
43	A	44	D	45	C	46	B	47	A	48	D
49	B	50	B	51	B	52	D	53	D	54	A
55	C	56	A	57	B	58	C	59	B	60	D
61	B	62	B	63	D	64	A	65	B	66	A